

# Sequential parametrized topological complexity of sphere bundles

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# Sectional Category

## Definition

Let  $p: E \rightarrow B$  be a fibration. The *sectional category*  $\text{secat}(p)$  of the fibration  $p$  is the minimum number  $k$  such that  $B$  can be covered by  $k + 1$  open subsets,

$$U_0 \cup U_1 \cup \cdots \cup U_k = B$$

and on each  $U_i$  there is a continuous section  $s_i: U_i \rightarrow E$  of  $p$ , i.e.,  $p \circ s_i = \text{Id}_{U_i}$ . If no such finite  $k$  exists then  $\text{secat}(p) = \infty$ .

## Lemma

For any fibration  $p: E \rightarrow B$  and coefficient ring  $R$ , if there exist cohomology classes  $u_1, \dots, u_k \in \ker[p^*: H^*(B; R) \rightarrow H^*(E; R)]$  such that their cup-product is nonzero,  $u_1 \cup \cdots \cup u_k \neq 0 \in H^*(B; R)$ , then  $\text{secat}(p) \geq k$ .

# Topological Complexity

- 1 Let  $X$  be a path connected topological space.
- 2  $PX = X^I$  the space of all continuous paths  $\gamma : [0, 1] \rightarrow X$ .
- 3 Consider the fibration  $p : PX \rightarrow X \times X$ , defined by  $p(\gamma) = (\gamma(0), \gamma(1))$  for any  $\gamma \in PX$ .

## Definition

The *topological complexity* (TC) of  $X$  is defined to be

$$TC(X) := \operatorname{secat}(p).$$

# Topological complexity

## Example

- 1  $TC(X) = 0$  if and only if  $X$  is contractible.
- 2  $TC(S^1) = 1$ . Since  $S^1$  is not contractible,  $TC(S^1) > 0$ . To show  $TC(S^1) = 1$ , we take

$$U_0 = \{(x, y) \in S^1 \times S^1; x \neq -y\}$$

$$U_1 = \{(x, y) \in S^1 \times S^1; x \neq y\}$$

Define  $s_0 : U_0 \rightarrow PS^1$  by  $s_0(x, y)$  is the unique shortest arc joining  $x$  and  $y$  and  $s_1 : U_1 \rightarrow PS^1$  by  $s_1(x, y)$  is the anticlockwise arc joining  $x$  and  $y$ .

- 3 For any  $n$ ,

$$TC(S^n) = \begin{cases} 1 & \text{if } n \text{ is odd} \\ 2 & \text{if } n \text{ is even} \end{cases}$$

# Sequential Topological Complexity

- 1 For any  $r \geq 2$  fix  $r$  points  $0 \leq t_1 < t_2 < \cdots < t_r \leq 1$  in the unit interval  $I$ .
- 2 Consider the fibration  $e_r : X^I \rightarrow X^r$ ,

$$\alpha \rightarrow (\alpha(t_1), \alpha(t_2), \alpha(t_3), \cdots, \alpha(t_r)),$$

where  $X^r = X \times X \times \cdots \times X$  and  $r \geq 2$ .

# Sequential Topological Complexity

## Definition

The  $r$ -th sequential topological complexity  $TC_r(X)$  of the space  $X$  is defined to be

$$TC_r(X) := \text{secat}(e_r).$$

Clearly,  $TC_2(X) = TC(X)$ .

## Remark

Let  $X$  be a path-connected space, then

- ①  $X$  is contractible if and only if  $TC_r(X) = 0$  for any  $r \geq 2$ .
- ②  $TC_r(X)$  is homotopy invariant.
- ③  $\{TC_r(X)\}_r$  is a non-decreasing sequence, that is  $TC_{r+1}(X) \geq TC_r(X)$  for any  $r \geq 2$ .

# Sequential Topological Complexity

## Example

For any  $n$ ,

$$TC_r(S^n) = \begin{cases} r - 1 & \text{if } n \text{ is odd} \\ r & \text{if } n \text{ is even} \end{cases}$$

# Bounds for Sequential Topological Complexity

## Proposition

We have the following relation:

$$\operatorname{cat}(X^{r-1}) \leq \operatorname{TC}_r(X) \leq \operatorname{cat}(X^r) \leq r \cdot \operatorname{cat}(X)$$

## Proposition

If  $X$  is  $k$ -connected CW complex then

$$\operatorname{TC}_r(X) < \frac{r \cdot \dim(X) + 1}{k + 1}.$$

# Bounds for Sequential Topological Complexity

- 1 We define the maps  $c : X \rightarrow X^I$  where  $c(x)(t) = x$  is the constant path. This map is a homotopy equivalence.
- 2 Consider the diagonal map  $\Delta : X \rightarrow X^r, x \mapsto (x, x, \dots, x)$ .
- 3 We have the following commutative diagram.

$$\begin{array}{ccc} X & \xrightarrow{c} & X^I \\ & \searrow \Delta & \swarrow e_r \\ & X^r & \end{array}$$

$$\ker[e_r^* : H^*(X^r; R) \rightarrow H^*(X^I; R)] = \ker[\Delta^* : H^*(X^r; R) \rightarrow H^*(X; R)]$$

# Sequential Parametrized Topological Complexity

- 1 Let  $p : E \rightarrow B$  be a Hurewicz fibration with fibre  $X$  and consider the space  $E_B^I \subset E^I$  the space of all paths  $\alpha : I \rightarrow E$  such that  $p \circ \alpha : I \rightarrow B$  is constant.

- 2 Fix an integer  $r \geq 2$  and denote

$$E_B^r = \{(e_1, \dots, e_r) \in E^r; p(e_1) = \dots = p(e_r)\}.$$

- 3 For  $r$  points  $0 \leq t_1 < t_2 < \dots < t_r \leq 1$  in the unit interval  $I$ .

- 4 Consider the evaluation map

$$\Pi_r : E_B^I \rightarrow E_B^r, \quad \Pi_r(\alpha) = (\alpha(t_1), \alpha(t_2), \dots, \alpha(t_r)).$$

## Definition

The  $r$ -th sequential parametrized topological complexity  $TC_r[p : E \rightarrow B]$  of the fibration  $p : E \rightarrow B$  is defined as,

$$TC_r[p : E \rightarrow B] := \text{secat}(\Pi_r).$$

# Sequential Parametrized Topological Complexity

- ① Let  $B' \subset B$  be a subset and let  $E' = p^{-1}(B')$  be its preimage, then obviously

$$TC_r[p : E \rightarrow B] \geq TC_r[p' : E' \rightarrow B']$$

where  $p' = p|_{E'}$ .

- ② In particular, taking  $B'$  to be a single point, we obtain

$$TC_r[p : E \rightarrow B] \geq TC_r(X),$$

where  $X$  is the fibre of  $p$ .

- ③ For a trivial fibration we get

$$TC_r[p : E \rightarrow B] = TC_r(X)$$

# Sequential Parametrized Topological Complexity

## Proposition

Let  $p : E \rightarrow B$  be a principal bundle with a connected topological group  $G$  as fibre. Then

$$TC_r[p : E \rightarrow B] = \text{cat}(G^{r-1}) = TC_r(G).$$

## Hint:

$$\begin{array}{ccc} P_0G \times E & \xrightarrow{F} & E_B^I \\ \downarrow \pi'_r \times \text{Id} & & \downarrow \Pi_r \\ G^{r-1} \times E & \xrightarrow{F'} & E_B^r \end{array}$$

where  $F : P_0G \times E \rightarrow E_B^I$  and  $F' : G^{r-1} \times E \rightarrow E_B^r$  are homeomorphisms given by

$$F(\alpha, x)(t) = \alpha(t)x \quad F'(g_2, g_3, \dots, g_r, x) = (x, g_2x, g_3x, \dots, g_rx).$$

## Example

Let  $p : S^3 \rightarrow S^2$  with fibre  $S^1$ . Then

$$TC_r[p : S^3 \rightarrow S^2] = TC_r(S^1) = r - 1$$

for any  $r \geq 2$ .

# Fibrewise Homotopy Invariance

## Remark

Let  $p : E \rightarrow B$  and  $p' : E' \rightarrow B$  be two fibrations and let  $f : E \rightarrow E'$  and  $g : E' \rightarrow E$  be two continuous maps such that the following diagram commutes

$$\begin{array}{ccc} & \xleftarrow{g} & \\ E & \xrightarrow{f} & E' \\ & \searrow p \quad \swarrow p' & \\ & B & \end{array}$$

If the map  $g \circ f : E \rightarrow E$  is fibrewise homotopic to the identity map then

$$TC_r[p : E \rightarrow B] \leq TC_r[p' : E' \rightarrow B].$$

# Some Properties

- ① Let  $p_1 : E_1 \rightarrow B$  and  $p_2 : E_2 \rightarrow B$  be two fibrations. Consider the fibration  $p : E \rightarrow B$  where  $E = E_1 \times_B E_2 = \{(e_1, e_2) \in E_1 \times E_2 \mid p_1(e_1) = p_2(e_2)\}$  and  $p(e_1, e_2) = p_1(e_1) = p_2(e_2)$ . Then

$$TC_r[p : E \rightarrow B] \leq TC_r[p_1 : E_1 \rightarrow B] + TC_r[p_2 : E_2 \rightarrow B].$$

## Proposition

Let  $p : E \rightarrow B$  be a locally trivial fibration with fiber  $X$ , where  $E, B, X$  are CW complexes. Assume that the fiber  $X$  is  $k$ -connected. Then

$$TC_r[p : E \rightarrow B] < \frac{hdim(E_B^r) + 1}{k + 1} \leq \frac{r \cdot \dim X + \dim B + 1}{k + 1}. \quad (1)$$

## Proposition

If there exist cohomology classes

$$u_1, \dots, u_k \in \ker[\Delta^* : H^*(E_B^r; R) \rightarrow H^*(E; R)]$$

such that  $u_1 \cup \dots \cup u_k \neq 0 \in H^*(E_B^r; R)$  then

$$TC_r[p : E \rightarrow B] \geq k.$$

# Sectional category of a sphere bundle

- 1  $\xi : E \rightarrow B$  rank  $q$  vector bundle, equipped with a metric structure.
- 2  $\xi : \dot{E} \rightarrow B$  associated bundle of  $(q - 1)$ -dimensional spheres. Here we estimate the sequential parametrized topological complexity  $TC_r[\dot{\xi} : \dot{E} \rightarrow B]$ .

## Definition (height)

For a cohomology class  $\alpha \neq 0$  denote by  $\mathfrak{h}(\alpha)$  its *height*, i.e. largest integer  $k$  such that its  $k$ -th power is nonzero,  $\alpha^k \neq 0$

## Theorem

- ① Sectional category of the sphere bundle fibration  $\dot{\xi} : \dot{E} \rightarrow B$  satisfies

$$\text{secat}(\dot{\xi}) \geq \mathfrak{h}(w_q(\xi))$$

where  $w_q(\xi) \in H^q(B; \mathbb{Z}_2)$  is the top Stiefel-Whitney class of  $\xi$ .

- ② If  $\xi$  is orientable then

$$\text{secat}(\dot{\xi}) \geq \mathfrak{h}(\mathfrak{e}(\xi))$$

- ③ If  $\xi$  is orientable and the base  $B$  is a CW-complex whose dimension satisfies

$$\dim B \leq q \cdot \mathfrak{h}(\mathfrak{e}(\xi)) + q$$

then

$$\text{secat}(\dot{\xi}) = \mathfrak{h}(\mathfrak{e}(\xi))$$

# Examples

## Example

Let  $\eta : E \rightarrow \mathbb{CP}^n$  be the canonical complex line bundle viewed as a rank 2 real vector bundle. Let  $\mathfrak{e}(\eta) \in H^2(\mathbb{CP}^n)$  be the generator and  $\mathfrak{h}(\mathfrak{e}(\eta)) = n$ . Since  $\dim \mathbb{CP}^n = 2n \leq 2 \cdot (n + 1)$ , it holds that  $\text{secat}(\eta) = n$ .

## Example

Let  $\xi_k = k\eta = \eta \oplus \eta \oplus \cdots \oplus \eta$  with  $rk(\xi_k) = 2k$ . Then  $\mathfrak{e}(\xi_k) = \mathfrak{e}(\eta)^k$  implying  $\mathfrak{h}(\mathfrak{e}(\xi_k)) = \lfloor n/k \rfloor$ . With dimension assumption it follows

$$\text{secat}(\xi_k) = \lfloor n/k \rfloor$$

## Example

Let  $\dot{\eta} : \dot{E}(\eta) \rightarrow \mathbb{CP}^n$  be the unit sphere bundle of the canonical complex line bundle over  $\mathbb{CP}^n$ . It is a principal  $S^1$ -bundle, its total space  $\dot{E}(\eta)$  is the sphere  $S^{2n+1}$ . Lower bound amounts to

$$TC(\dot{\eta} : \dot{E}(\eta) \rightarrow \mathbb{CP}^n) \geq TC(S^1) = 1$$

To obtain the inverse inequality we consider the following partition

$$\dot{E}(\eta) \times_B \dot{E}(\eta) = F_0 \amalg F_1$$

where  $F_0$  is the set of all pairs  $(z_1, z_2)$  of unit vectors  $z_1, z_2 \in S^{2n+1}$  lying in the same fibre but not antipodal, i.e.  $z_1 \neq -z_2$ ;  $F_1$  is the set of antipodal pairs  $(z, -z)$ . For  $(z_1, z_2) \in F_0$  we can write  $z_2/z_1 = e^{i\phi}$  where  $\phi \in (-\pi, \pi)$  and a continuous section  $s_0 : F_0 \rightarrow E_B^I$  can be defined as

$$s_0(z_1, z_2)(t) = e^{i\phi t} z_1, \quad t \in [0, 1]$$

## Example

On the other hand, over  $F_1$  we can define a continuous section  $s_1$  where

$$s_1(z, -z)(t) = e^{i\pi t} z \quad t \in [0, 1]$$

So the result is

$$TC[\dot{\eta} : \dot{E}(\eta) \rightarrow \mathbb{CP}^n] = 1$$

# Leray-Hirsch theorem

## Theorem

Let  $F \rightarrow E \xrightarrow{p} B$  be a fiber bundle such that for some commutative coefficient ring  $R$ :

- 1  $H^n(F; R)$  is a finitely generated free  $R$ -module for each  $n$
- 2 There exist classes  $c_j \in H^{k_j}(E; R)$  whose restrictions  $i^*(c_j)$  form a basis for  $H^*(F; R)$  in each fiber  $F$ , where  $i : F \rightarrow E$  is the inclusion.

Then the map  $\Phi : H^*(B; R) \otimes_R H^*(F; R) \rightarrow H^*(E; R)$  defined as

$$\sum_{ij} b_i \otimes i^*(c_j) \mapsto \sum_{ij} p^*(b_i) \cup c_j$$

is an isomorphism of  $R$ -modules.

$H^*(E; R)$  is a free  $H^*(B; R)$ -module with basis  $\{c_j\}$ . We do not get a ring isomorphism but only isomorphism of abelian groups.

# The cup-length associated with a section

- 1  $\dot{\xi} : \dot{E} \rightarrow B$  admits cohomological extension of the fibre iff  $\mathfrak{e}(\xi) = 0$ , hence satisfies Leray-Hirsch theorem.
- 2 Denote  $\dot{F} = \{e \in \dot{E}; e \perp s(\xi(e))\}$  and  $\eta = \xi|_{\dot{F}}$  an oriented bundle of  $(q-2)$ -spheres.

## Theorem

$$\text{cupker}[s^* : H^*(\dot{E}, \mathbb{Z}) \rightarrow H^*(B, \mathbb{Z})] = \mathfrak{h}(\mathfrak{e}(\eta)) + 1$$

If we use  $\mathbb{Z}_2$  coefficients the role of the Euler class plays the top Stiefel-Whitney class of the bundle  $\eta$  of vectors orthogonal to the section. The advantage of the following corollary is that the answer is given in terms of the original bundle  $\xi$  and its characteristic class  $w_{d-1}(\xi) \in H^{q-1}(B, \mathbb{Z}_2)$ .

## Corollary

Let  $\xi : E \rightarrow B$  be a rank  $q \geq 2$  vector bundle (not necessarily orientable). Let  $s : B \rightarrow \dot{E}$  be a continuous section of the unit sphere bundle. Then the cup-length of the kernel  $\ker[s^* : H^*(\dot{E}, \mathbb{Z}_2) \rightarrow H^*(B; \mathbb{Z}_2)]$  equals  $\mathfrak{h}(w_{q-1}(\xi)) + 1$

# Parametrized topological complexity of sphere bundles

## Definition

$\ddot{\xi} : \ddot{E} \rightarrow \dot{E}; (x, y) \mapsto x$  where  $\ddot{E} \subset \dot{E} \times_B \dot{E}$  is the space of pairs of mutually orthogonal unit vectors.

$\ddot{\xi}$  is an oriented locally trivial fibration with fibres  $S^{q-2}$ .

The Euler class  $e(\ddot{\xi}) \in H^{q-1}(\dot{E})$  has an obvious property that for any point  $b \in B$  the restriction  $e(\ddot{\xi})|_{\dot{E}_b}$  is the Euler class of the tangent bundle of the sphere  $\dot{E}_b$ .

## Remark

If  $\xi$  admits a complex structure,  $secat(\ddot{\xi}) = 0$ , section being defined as  $s(e) = \sqrt{-1} \cdot e$ .

# Estimates

## Theorem

*The following estimates hold*

$$\mathfrak{h}(\mathfrak{e}(\ddot{\xi})) + 1 \leq TC[\dot{\xi} : \dot{E} \rightarrow B] \leq \text{secat}(\ddot{\xi}) + 1$$

*Moreover, if  $B$  is a CW-complex satisfying  $\dim B \leq (q - 1) \cdot \mathfrak{h}(\mathfrak{e}(\ddot{\xi}))$  then*

$$\mathfrak{h}(\mathfrak{e}(\ddot{\xi})) + 1 = TC[\dot{\xi} : \dot{E} \rightarrow B] = \text{secat}(\ddot{\xi}) + 1$$

## Corollary

For a vector bundle  $\xi : E \rightarrow B$  satisfying  $\text{secat}(\ddot{\xi}) = 0$  (i.e. it admits a complex structure) one has  $TC[\dot{\xi}] = 1$  or

$$TC_r[\dot{\xi}] = r - 1 \tag{2}$$

in sequential case.

For  $q$  even the Euler class  $\mathfrak{e}(\ddot{\xi}) \in H^{q-1}(\dot{E})$  has order 2, i.e.  $2 \cdot \mathfrak{e}(\ddot{\xi}) = 0$ . Let us focus on the case when  $q$  is odd. The corollary below has the advantage of dealing with cohomology of the base  $B$ .

## Corollary

For  $q \geq 3$  odd let  $\eta : E(\eta) \rightarrow B$  be an oriented vector bundle of rank  $q - 1$ . Let  $\xi = \eta \oplus \epsilon$  be the sum where  $\epsilon$  is the trivial line bundle over  $B$ . Then one has

$$TC[\dot{\xi} : \dot{E}(\xi) \rightarrow B] \geq \mathfrak{h}(\mathfrak{e}(\eta)) + 1$$

Moreover, if height  $\mathfrak{h}(\mathfrak{e}(\eta))$  is even and the integral cohomology of the base  $B$  in dimension  $(q - 1) \cdot \mathfrak{h}(\mathfrak{e}(\eta))$  has no 2-torsion then

$$TC[\dot{\xi} : \dot{E}(\xi) \rightarrow B] \geq \mathfrak{h}(\mathfrak{e}(\eta)) + 2$$

## Example

Denote  $\xi = \eta \oplus \epsilon$ , where  $\eta : E(\eta) \rightarrow \mathbb{CP}^n$  is the canonical bundle and  $\epsilon$  a trivial bundle. We have  $rk(\xi) = 3$  odd and  $\mathfrak{h}(\mathfrak{e}(\eta)) = n$ . By the previous corollary we get  $TC[\dot{\xi} : \dot{E}(\xi) \rightarrow \mathbb{CP}^n] \geq n + 1$  and moreover for  $n$  even  $TC[\dot{\xi} : \dot{E}(\xi) \rightarrow \mathbb{CP}^n] \geq n + 2$ . With the upper bound, we see

$$TC[\dot{\xi} : \dot{E}(\xi) \rightarrow \mathbb{CP}^n] = n + 2$$

for all even  $n$ . In particular we see that the parametrized TC of sphere bundles can be arbitrary large. This contrasts the situation with the unparametrized TC which takes the value of 1 and 2 only for spheres.

# Upper bound for sphere bundle

## Proposition

Let the vector bundle  $\xi : E(\xi) \rightarrow B$  be the Whitney sum  $\xi = \xi_1 \oplus \xi_2$ .  
Then

$$\operatorname{secat}[\ddot{\xi}] \leq \operatorname{secat}[\dot{\xi}_1] + \operatorname{secat}[\dot{\xi}_2] + 1$$

## Corollary

For the above case  $TC_r[\dot{\xi}] \leq \operatorname{secat}[\dot{\xi}_1] + \operatorname{secat}[\dot{\xi}_2] + r$

# Upper bound for sphere bundle

## Corollary

If  $\xi = \eta \oplus \epsilon$  where  $\epsilon$  is the trivial line bundle then

$$TC_r[\dot{\xi}] \leq \text{secat}[\dot{\eta}] + r.$$

## Corollary

If a vector bundle  $\xi$  admits two non-vanishing linearly independent sections then

$$TC_r[\dot{\xi}] \leq r.$$

If additionally the rank  $q$  of  $\xi$  is odd then,

$$TC_r[\dot{\xi}] = r. \tag{3}$$

Indeed, for the second case we have  $r \geq TC_r[\dot{\xi}] \geq TC_r(S^{q-1}) = r$ .

# A lower bound for sphere bundle

## Theorem

Consider the space  $\dot{E}_B^r$ , where  $r \geq 2$ , and the diagonal map  $\Delta : \dot{E} \rightarrow \dot{E}_B^r$ , where  $\Delta(e) = (e, e, \dots, e)$  for  $e \in \dot{E}$ . Then the cup-length of the kernel

$$\ker[\Delta^* : H^*(\dot{E}_B^r, \mathbb{Z}) \rightarrow H^*(\dot{E}, \mathbb{Z})]$$

equals  $\mathfrak{h}(\mathfrak{e}(\ddot{\xi})) + r - 1$ . Here  $\mathfrak{h}(\mathfrak{e}(\ddot{\xi}))$  is the largest number  $k$  such that  $\mathfrak{e}(\ddot{\xi})^k \in H^*(\dot{E}, \mathbb{Z})$  is nonzero.

## Corollary

The sequential parametrized topological complexity of the unit sphere bundle associated with an oriented vector bundle  $\xi : E \rightarrow B$  is bounded below by

$$TC_r[\dot{\xi}] \geq \mathfrak{h}(\mathfrak{e}(\ddot{\xi})) + r - 1.$$

So we get

$$\mathfrak{h}(\mathfrak{e}(\ddot{\xi})) + r - 1 \leq TC_r[\dot{\xi}] \leq \text{secat}[\ddot{\xi} : \ddot{E} \rightarrow \dot{E}] + r - 1.$$

# Sphere bundle

## Corollary

Let  $\xi : E \rightarrow B$  be an orientable rank  $q$  vector bundle where the base  $B$  is a CW-complex of dimension

$$\dim B \leq (q - 1) \cdot \mathfrak{h}(\mathfrak{e}(\ddot{\xi})).$$

Then for any  $r \geq 2$  one has

$$TC_r[\dot{\xi}] = \mathfrak{h}(\mathfrak{e}(\ddot{\xi})) + r - 1 = \text{secat}[\ddot{\xi} : \ddot{E} \rightarrow \dot{E}] + r - 1.$$

**Hint:** Here we use the following result: If  $\eta$  is an orientable vector bundle of dimension  $q'$  and the base  $B'$  is a CW-complex with

$$\dim B' \leq q' \cdot (\mathfrak{h}(\mathfrak{e}(\dot{\eta})) + 1)$$

then,

$$\text{secat}(\dot{\eta}) = \mathfrak{h}(\mathfrak{e}(\dot{\eta})).$$

We apply the above result to the bundle  $\zeta : E(\zeta) \rightarrow \dot{E}$ , where

$$E(\zeta) = \{(e, e') \in \dot{E}(\xi) \times_B E(\xi); e \perp e'\}$$

and  $\zeta(e, e') = e$ . Then  $\zeta$  is  $q - 1$  dimensional vector bundle and its associated sphere bundle is  $\ddot{\xi} : \ddot{E} \rightarrow \dot{E}$ . Here we see that  $\dim \dot{E} \leq \dim B + q - 1 \leq (q - 1)(\mathfrak{h}(\mathfrak{e}(\ddot{\xi})) + 1)$ . Applying above result on  $\zeta$  we get

$$\text{secat}[\ddot{\zeta} : \ddot{E} \rightarrow \dot{E}] = \mathfrak{h}(\mathfrak{e}(\ddot{\xi})).$$

## Proposition

Assume that  $\xi = \eta \epsilon$  where  $\eta : E(\eta) \rightarrow B$  is an oriented rank  $q - 1$  vector bundle and  $\epsilon$  is the trivial oriented line bundle. Then

$$TC_r[\dot{\xi}] \geq \mathfrak{h}(\epsilon(\eta)) + r - 1. \quad (4)$$

Moreover, if (i)  $q \geq 3$  is odd, (ii)  $\mathfrak{h}(\epsilon(\eta))$  is even and (iii)  $H^*(B; \mathbb{Z})$  has no 2-torsion, then

$$TC_r[\dot{\xi}] \geq \mathfrak{h}(\epsilon(\eta)) + r. \quad (5)$$

## Another upper bound

### Proposition

Let  $\xi : E \rightarrow B$  be an oriented rank  $q \geq 3$  vector bundle. Suppose that  $B$  is a simply connected finite CW-complex and its dimension  $\dim B$  is divisible by  $q - 1$  and satisfies

$$\mathfrak{h}(\mathfrak{e}(\ddot{\xi})) \leq \frac{\dim B}{q - 1}.$$

Then

$$TC_r[\dot{\xi}] \leq r - 1 + \frac{\dim B}{q - 1}. \quad (6)$$

This upper bound is by one better than the previous one 1.

# Sphere bundle

## Example

Consider the canonical complex line bundle  $\eta : E(\eta) \rightarrow \mathbb{CP}^n$  over the complex projective space. Consider the bundle

$$\xi = \eta \oplus \epsilon$$

of rank  $q = 3$  and here  $\mathfrak{h}(\epsilon(\eta)) = n$ . Using lower bounds 4 and 5 we obtain

$$TC_r[\dot{\xi} : \dot{E}(\xi) \rightarrow \mathbb{CP}^n] \geq \begin{cases} n + r & \text{if } n \text{ is even} \\ n + r - 1 & \text{if } n \text{ is odd.} \end{cases}$$

Now using the upper bound 6 and 1 we get

$$TC_r[\dot{\xi} : \dot{E}(\xi) \rightarrow \mathbb{CP}^n] = \begin{cases} n + r & \text{if } n \text{ is even} \\ n + r - 1 & \text{if } n \text{ is odd.} \end{cases}$$

# Sphere bundle

## Example

Consider the more general case

$$\xi_l = \eta \oplus l \cdot \epsilon = \eta \oplus \epsilon \oplus \epsilon \oplus \cdots \oplus \epsilon$$

where  $\eta : E(\eta) \rightarrow \mathbb{CP}^n$  is the canonical complex line bundle,  $l = 0, 1, 2, \dots$ . The rank of  $\xi_l$  equals  $q = 2 + l$ . The previous example is discussed for  $l = 1$ .

- 1 If  $l$  is odd and  $l > 1$  then  $\xi_l$  admits two linear independent sections and hence by 3 we obtain  $TC_r[\dot{\xi}] = r$ .
- 2 If  $l$  is even then  $\xi_l$  admits a complex structure and hence  $TC_r[\dot{\xi}] = r - 1$ , using 2

# Lower bound using Stiefel-Whitney classes

## Theorem

*Let  $\xi : E \rightarrow B$  be a locally trivial vector bundle of rank  $q \geq 2$ . Consider the associated sphere bundle  $\dot{\xi} : \dot{E} \rightarrow B$ . Then*

$$TC_r[\dot{\xi}] \geq \mathfrak{h}(w_{q-1}(\xi)|w_q(\xi)) + r - 1. \quad (7)$$

Here  $w_{q-1}(\xi) \in H^{q-1}(B; \mathbb{Z}_2)$  is an Stiefel-Whitney class and  $\mathfrak{h}(w_{q-1}(\xi)|w_q(\xi))$  denotes the smallest integer  $k \geq 0$  such that the power  $w_{q-1}(\xi)^{k+1}$  lies in the ideal of the algebra  $H^*(B; \mathbb{Z}_2)$  generated by the class  $w_q(\xi)$ .

# Example

- 1 Consider the bundle

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- 6 So,  $\mathfrak{h}(w_{q+1}(\xi_l)) = \lceil \frac{n+1-l}{l} \rceil = \lceil \frac{n+1}{l} \rceil - 1$ .

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① Using 7 we obtain

$$TC_r[\dot{\xi}_l : \dot{E}(\xi_l) \rightarrow \mathbb{RP}^n] \geq \lceil \frac{n+1}{l} \rceil + r - 2.$$

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For the above case we have

$$\lceil \frac{n+1}{l} \rceil + r - 2 \leq TC_r[\dot{\xi}_l] \leq \lceil \frac{n+1}{l} \rceil + r - 1$$

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# Thanks!

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